

# Generalized Estimating Equations

A plain-language guide to fitting models for repeated and clustered measurements — when you only need to trust your guess about the average pattern, not your guess about how correlated the data are.

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Based on the chapter by Geert Molenberghs, Geert Verbeke, and Michael G. Kenward, in the Handbook of Epidemiology (Springer)

## Two Ways to Ask the Same Question

When the same people are measured more than once, or grouped in clusters, a model's coefficients can be read in two different ways: how does an exposure change the population's average outcome, or how does it change a given person's own outcome? Epidemiology is usually interested in the first.

### **Marginal (Population-Averaged)**

How does the average outcome across the whole population shift with an exposure or treatment, holding confounders fixed? This is what a policy question or a treatment comparison usually wants.

### **Subject-Specific**

How does this particular person's own outcome change, given their own past values? This matters more for predicting one person's future course, as with an empirical Bayes prediction.

For ordinary (Gaussian) continuous outcomes, one flexible tool — the linear mixed model (Laird and Ware, 1982) — conveniently supports both readings at once. That convenience does not carry over automatically once outcomes are binary, counts, or time-to-event.

## Why Non-Gaussian Outcomes Make This Hard

Once outcomes are binary, counts, or time-to-event rather than continuous, the convenient dual interpretation breaks down. A model built for a clean hierarchical, subject-specific interpretation — a generalized linear mixed model, with random effects folded into a logit or similar link — generally does not also have a closed-form marginal, population-averaged interpretation. You cannot easily have both.

Fully parametric models built explicitly for a marginal mean interpretation do exist — such as the Bahadur (1961) model or the multivariate odds ratio model — but they become computationally unworkable quickly: the number of parameters grows exponentially with the number of repeated measurements, and the valid range for the association parameters is often severely restricted.

What is needed instead is a method for the common situation where the real interest lies in the marginal mean, and perhaps simple pairwise associations — without insisting on a fully specified joint distribution for the whole sequence of measurements.

## A Shortcut: You Don't Need the Whole Distribution

Wedderburn (1974) showed something quietly radical for independent data: to estimate regression parameters well, you do not need to specify the full probability distribution of the outcome. You only need to get two things right — the mean, and how the variance relates to that mean (a “variance function”), possibly up to an unknown scaling factor.

Estimating equations built only from a mean and a variance function are called **quasi-likelihood**. Wedderburn showed that for the exponential family of distributions — which covers Gaussian, binomial, and Poisson outcomes — quasi-likelihood produces the same estimates as the true likelihood, even when a link function other than the “natural” one is used.

This is the seed idea behind everything that follows: a correctly specified mean, plus a plausible variance assumption, can carry an enormous amount of the statistical weight on its own.

## Extending the Shortcut to Correlated Data

Liang and Zeger (1986) took Wedderburn's quasi-likelihood idea — built for independent, cross-sectional data — and extended it to longitudinal, clustered, or otherwise hierarchical data, where the same person or unit contributes several correlated measurements. They named the result **generalized estimating equations**, or GEE (sometimes called GEE1, to distinguish it from later extensions).

Their requirement: you only need to correctly specify the marginal distribution of each individual measurement — its own mean and variance — provided you are willing to adopt a working assumption about how measurements within the same person or cluster relate to one another.

## The Big Idea: A Guess You're Allowed to Get Wrong

That working assumption is called a **working correlation**. The word “working” is doing real labor: it signals that this piece of the model is allowed to be incorrect. Liang and Zeger’s central theoretical result — explored in depth later in this deck — is that as long as the mean model itself is correctly specified, GEE’s estimates of the regression coefficients stay correct on average even when the working correlation guess is wrong.

You may lose some statistical efficiency — precision — from a poor guess, but not validity. This is what makes GEE so practically appealing: an analyst does not need to know the true dependence structure of repeated or clustered measurements to get trustworthy answers about an exposure-outcome relationship.

## THE TOENAIL TRIAL, 1 OF 4

# A Trial for a Common, Stubborn Infection

De Backer et al. (1996) ran a randomized, double-blind, parallel-group, multicenter trial comparing two oral antifungal treatments — labeled A and B — for toenail dermatophyte onychomycosis (TDO), a common infection affecting more than 2% of the population that, once established, needs treatment until the whole nail has grown out healthy. The trial compared 12 weeks of continuous therapy with one treatment or the other, across 36 centers.

378 patients ( $2 \times 189$ ) were randomized overall, followed through 12 weeks of treatment and further to a total of 48 weeks (12 months), with visits at baseline, monthly during treatment, and every 3 months afterward — up to 7 occasions per patient.

At the first visit, physicians identified one affected “target nail” per patient to track over time. This chapter’s analyses — and this deck’s — follow the 146 Group A and 148 Group B patients whose target nail was a big toenail.

THE TOENAIL TRIAL, 2 OF 4

# Did Severe Infections Decline the Same Way in Both Arms?

The outcome tracked at every visit was simply whether the target nail’s infection was severe (yes or no). The question of interest: did the percentage of severe infections fall similarly under both treatments, or did one pull ahead?

| Occasion  | Group A % severe (N) | Group B % severe (N) |
|-----------|----------------------|----------------------|
| Baseline  | 37.0% (146)          | 37.2% (148)          |
| 1 month   | 34.7% (141)          | 32.6% (147)          |
| 2 months  | 31.9% (138)          | 27.6% (145)          |
| 3 months  | 22.0% (132)          | 20.7% (140)          |
| 6 months  | 10.8% (130)          | 6.0% (133)           |
| 9 months  | 8.5% (117)           | 6.3% (127)           |
| 12 months | 10.5% (133)          | 4.6% (131)           |

Both arms improved substantially, but Group B’s decline looks somewhat steeper from 6 months onward — exactly the kind of gap a formal treatment-by-time interaction test is built to check.

## Not Everyone Has All Seven Measurements

A common reality of longitudinal studies: not every patient completes every scheduled visit. Of the 298 patients with a big-toenail target, only 224 (76%) were measured at all 7 scheduled time points; the rest dropped out or missed visits at various points along the way.

**Group A**

107 of 146 patients (73.29%) had all 7 measurements; the remaining 39 patients had anywhere from 1 to 6.

**Group B**

117 of 148 patients (79.05%) had all 7 measurements; the remaining 31 patients had anywhere from 1 to 6.

This kind of unequal, “unbalanced” follow-up is exactly the situation a marginal, GEE-based approach is built to accommodate gracefully — without discarding anyone’s partial data.

GEE can use the available measurements, but standard validity under dropout still requires strong assumptions about why measurements are missing; we return to this later.

## THE TOENAIL TRIAL, 4 OF 4

# Translating the Question Into a Model

At each visit, the outcome is simply whether that visit found a severe infection — a yes/no outcome. The model expresses the chance of a severe result as a function of three things: which treatment group the patient is in, how much time has passed since baseline, and the interaction between the two.

That interaction term is the one of real scientific interest: it asks whether the two treatments' rates of improvement over time actually differ, rather than whether the groups simply started out different (which good randomization should rule out).

All of the original analyses were run in SAS: PROC GENMOD (or equivalently PROC GEE) for standard GEE and for alternating logistic regression, and PROC GLIMMIX for the linearization-based method introduced later in this deck.

## What the Estimating Equation Actually Does

For each person, GEE compares their observed sequence of measurements to what the marginal mean model predicts for them. It weighs that mismatch by an assumed covariance — built from the variance function (how spread out each measurement is) and the working correlation (how the measurements within that person relate to one another) — and adds these weighted mismatches up across everyone in the study.

The regression coefficients are then chosen to drive that total weighted mismatch to zero — the same logic as ordinary weighted least squares, just built from a working guess about correlation rather than the true one.

## Why a Working Correlation Is Needed At All

The marginal mean model, on its own, says nothing about how a person's repeated measurements relate to one another. That has to come from somewhere else — an extra piece of the model, governed by its own extra parameters, describing the correlation among measurements within the same person or cluster.

Why not just model that correlation carefully and fully? Because doing so honestly would drag in the third and fourth moments of the data too, ultimately requiring the entire joint distribution of the repeated measurements — exactly the complexity GEE was designed to sidestep.

Liang and Zeger's way out: let the analyst specify a plausible working correlation structure, without demanding it be exactly right.

# Five Common Shapes for the Working Guess

| Structure    | What It Assumes                                                                                                                               | Best Suited For                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Independence | No correlation at all between repeated measurements — the simplest possible guess                                                             | A starting point, or when correlation is expected to be minimal |
| Exchangeable | Every pair of measurements on the same person is equally correlated, no matter how far apart in time                                          | Clustered data with no natural ordering, like littermates       |
| AR(1)        | Correlation fades with time apart, following one steady, fixed rate of decay                                                                  | Evenly spaced repeated measures with a clear time order         |
| Toeplitz     | Correlation depends on time apart, like AR(1), but each specific gap gets its own separately estimated value rather than one fixed decay rate | Ordered measures where decay may not be perfectly steady        |
| Unstructured | No pattern assumed at all — a separate correlation estimated for every single pair of occasions                                               | Few, common time points and a large enough sample               |

AR(1), Toeplitz, and unstructured structures are most meaningful when repeated measurements can be consistently ordered or aligned across clusters. For unordered clusters such as littermates, independence or exchangeable working correlations are usually the natural choices.

## The Theorem That Makes GEE Work

Liang and Zeger proved that, provided the marginal mean model is correctly specified, and under mild regularity conditions, the GEE estimator of the regression coefficients is consistent — it converges to the true values as the sample grows — and approximately normally distributed, regardless of whether the working correlation matrix is the true one.

If the working correlation happens to be exactly right, GEE reaches full efficiency, matching what a correctly specified likelihood would give. If it is wrong, there is no bias — only a possible loss of precision, because the estimator is no longer weighting each person's information in the optimal way.

There is no price to pay for correctly specifying the working correlation; there may be an efficiency cost for getting it badly wrong. Either way, the estimates themselves stay trustworthy.

## Two Ways to Compute Standard Errors

Because the working correlation might be wrong, GEE offers two different recipes for standard errors, confidence intervals, and p-values.

### Naive (Model-Based)

Assumes the working correlation is exactly correct. If it isn't, these standard errors can be too small — producing overconfident, falsely significant results.

### Robust (Sandwich, Empirically Corrected)

Estimated directly from the observed data patterns. Stays trustworthy whether or not the working correlation guess was right.

The recommendation: base inferences on the robust, empirically corrected standard errors. Comparing the two also works as an informal diagnostic — when they disagree sharply, that is a sign the working correlation was a poor choice.

## LIANG AND ZEGER'S GEE, 6 OF 6

# Fitting GEE: A Back-and-Forth Loop

Since the coefficient estimates and the working correlation estimate each depend on the other, GEE is fit by alternating between them until both settle down.

1. Start simple: get initial coefficient estimates by ignoring correlation entirely (ordinary logistic regression for binary data, Poisson regression for counts).
2. Use the leftover mismatches from that starting model to estimate the working correlation parameters, and a scale (overdispersion) parameter if needed.
3. Combine the mean model and the estimated working correlation into a covariance estimate for each person.
4. Update the coefficient estimates using that covariance estimate, much like a weighted regression step.

Steps 2 through 4 repeat until the estimates stop changing meaningfully — that is, until the algorithm converges.

APPLYING GEE TO THE TOENAIL DATA, 1 OF 3

Five Working Guesses, One Trial

Fitting the same treatment-by-time model to the toenail data under five different working correlations gives very similar point estimates for the treatment-by-time interaction — but the two flavors of p-value for it tell noticeably different stories.

| Working Correlation | Interaction Estimate | p-value (naive; robust) | QIC     |
|---------------------|----------------------|-------------------------|---------|
| Independence        | -0.078               | 0.0470; 0.1515          | 1833.34 |
| Exchangeable        | -0.089               | 0.0143; 0.1208          | 1834.38 |
| AR(1)               | -0.105               | 0.0277; 0.0458          | 1834.18 |
| Toeplitz            | -0.102               | 0.0301; 0.0513          | 1833.97 |
| Unstructured        | -0.114               | 0.0158; 0.0285          | 1836.26 |

All five agree the interaction is negative — Group B improving faster — but the robust p-values range from a clearly significant 0.0285 to a much less convincing 0.1515.

## APPLYING GEE TO THE TOENAIL DATA, 2 OF 3

### A Cautionary Tale in the Numbers

Under independence, the naive p-value for the treatment effect is 0.0470 — just under the conventional 0.05 threshold, suggesting a significant finding. But the robust p-value for the very same model is 0.1515 — not significant at all. The exchangeable structure shows the same pattern: 0.0143 naive versus 0.1208 robust.

In every one of the five working correlations tried here, the model-based p-value would lead to an overly optimistic conclusion about the treatment effect. Independence and exchangeable are the least efficient choices for this data, exactly as shown by how much their naive and robust p-values disagree.

## APPLYING GEE TO THE TOENAIL DATA, 3 OF 3

# A Working Model Can Still Be Used Pragmatically

One could object to using AR(1) or Toeplitz here at all: the toenail visits are not evenly spaced — monthly for the first quarter, then every three months afterward — which is exactly what those two structures assume.

In practice, though, both still produced conclusions close to the fully flexible unstructured model. Because the working correlation is only ever a working model — never a claim about the truth — a pragmatic analyst can use a structure like this anyway, so long as the robust standard errors are what inference relies on.

The larger lesson: choice of working correlation does not affect whether GEE estimates are trustworthy, only how much precision you get and how cautious the naive standard errors are likely to be.

## REFINEMENTS TO GEE, 1 OF 5

# Prentice: Give the Correlation Its Own Equation

Prentice (1988) offered an alternative to Liang and Zeger's simple, moment-based estimate of the working correlation. Rather than plugging in a quick estimate, Prentice supplemented the mean's estimating equation with a second, separate estimating equation built specifically for the correlation parameters.

The payoff: formal precision estimates (standard errors) for the correlation parameters themselves, not just the mean parameters — a genuine increase in what can be learned, at some cost in modeling effort. Lipsitz, Laird, and Harrington (1991) later modified this approach to use pairwise odds ratios in place of correlations.

The two estimating equations are typically treated as independent of one another. This can sacrifice a little statistical efficiency, but it helps preserve the robustness of the marginal regression estimates even if the association model itself is misspecified.

## REFINEMENTS TO GEE, 2 OF 5

# GEE2: Estimating the Mean and the Association Together

Second-order generalized estimating equations (GEE2) — proposed by Zhao and Prentice (1990) using correlations, and by Liang, Zeger, and Qaqish (1992) using odds ratios — go a step further than Prentice's method: instead of treating the mean and the association as two separate, independently estimated pieces, GEE2 folds them into one joint estimating equation.

This is more efficient — it uses the available information more fully — but considerably more demanding computationally, since it requires working with third- and fourth-order association quantities. These are typically just set to zero, a working assumption in the same spirit as the working correlation, one level up.

The same principle can, in theory, be extended to even higher-order estimating equations — but the added complexity quickly outweighs the scientific value of information that high-order associations rarely provide.

## A More Natural Measure for Binary Data: Odds Ratios

Several authors have pointed out that for binary or categorical outcomes, the odds ratio is a more natural measure of association between two repeated measurements than a correlation coefficient — it does not run into the awkward range restrictions that correlations can face once outcomes are binary.

Lipsitz, Laird, and Harrington (1991) used odds ratios within Prentice's two-equation framework; Liang, Zeger, and Qaqish (1992) did the same within GEE2. Both still need a working assumption — usually setting the third- and fourth-order log odds ratios to zero — to keep the computation tractable.

## REFINEMENTS TO GEE, 4 OF 5

# Alternating Logistic Regression: The Best of Both Worlds

Carey, Zeger, and Diggle (1993) proposed **alternating logistic regression** (ALR) — a genuinely different approach that borrows strengths from both GEE1 and GEE2. It fits two logistic models by turns: an ordinary one for the marginal mean, and a second one for the chance that one measurement is positive given the value of another measurement on the same person, with the log odds ratio between them as the association parameter.

The two models are fit by iterating between them. The advantage that sets ALR apart: it delivers precision estimates for both the mean and the association parameters, is nearly as efficient as GEE2, is computationally about as easy as GEE1 — and, unlike GEE2, needs absolutely no working assumption about third- or fourth-order associations at all. It sidesteps that problem entirely, rather than just assuming it away.

ALR is specifically a binary-outcome, marginal-logit method — not a general replacement for GEE across every outcome type.

ALR's log odds ratios can follow flexible structures of their own — an “exchangeable” equivalent, a fully unstructured set at common time points, or banded structures under equal spacing. Heagerty and Zeger (1996) later offered a related general method, though it still needs third- and fourth-order probabilities, just computed more easily than in full GEE2.

# ALR on the Toenail Data

ALR was run with three correlation structures. Under independence, the results are identical to standard GEE. AR(1) is not available for ALR, since it is tied specifically to correlations rather than odds ratios; a Toeplitz structure was built by hand, treating pairs of visits the same number of occasions apart as sharing one odds ratio; an unstructured version exists in principle but failed to converge for this data.

| Structure    | Treatment-by-Time Estimate | p-value (naive; robust) |
|--------------|----------------------------|-------------------------|
| Independence | -0.078 (same as GEE1)      | 0.0470; 0.1515          |
| Exchangeable | -0.084                     | 0.0324; 0.1076          |
| Toeplitz     | -0.097                     | 0.0331; 0.0539          |

The Toeplitz odds ratios themselves fall off sharply and steadily with distance — from about 4.8 for visits one occasion apart down to about 1.0 for visits six occasions apart — a clear, interpretable signature of fading association over time. Overall, the available structures point to conclusions similar to their GEE1 counterparts.

## AN ALTERNATIVE ROUTE, 1 OF 2

# A Different Computational Path to a Similar Place

Every GEE version so far builds a multivariate extension of a generalized linear model's own scoring equations, adjusted for association. An alternative strategy instead linearizes the outcome: it builds a “working variate” from a simple straight-line approximation around the current fitted mean, then applies weighted least squares to that working variate.

This is **iteratively reweighted least squares** (IRLS) — the same technique used to fit an ordinary generalized linear model, extended here to correlated, multivariate data. Each step produces the same kind of quantities found in a standard multivariate normal model, so familiar multivariate regression software — such as SAS's PROC GLIMMIX — can be used directly.

AN ALTERNATIVE ROUTE, 2 OF 2

A Different Path, a Similar Destination

Refitting the same treatment-by-time model with the linearization-based method, across the same five working correlations, gives results in close qualitative agreement with standard GEE — with a slight gain in precision.

| Working Correlation | Interaction Estimate | p-value (naive; robust) |
|---------------------|----------------------|-------------------------|
| Independence        | -0.078               | 0.525; 0.1517           |
| Exchangeable        | -0.089               | 0.0193; 0.1206          |
| AR(1)               | -0.125               | 0.0115; 0.0262          |
| Toeplitz            | -0.131               | 0.0096; 0.0221          |
| Unstructured        | -0.106               | 0.0120; 0.0170          |

## CHOOSING AND CHECKING A MODEL, 1 OF 3

# Judging Fit Without a True Likelihood

Full maximum likelihood offers familiar model-selection tools: Wald, score, and likelihood ratio tests. GEE, lacking a true likelihood basis, routinely offers only Wald-type statistics, which need just parameter and precision estimates. Rotnitzky and Jewell (1990) built GEE versions of score and likelihood ratio tests, later adapted into pseudo-likelihood form by Geys et al. (1999).

For comparing whole models — different predictors, or different working correlations — Pan (2001) proposed **QIC**, a GEE analog of Akaike's Information Criterion, built from the independence-based quasi-likelihood and corrected using the robust variance-covariance matrix.

CHOOSING AND CHECKING A MODEL, 2 OF 3

# What QIC Says About the Toenail Models

Lower QIC points to a more adequate working correlation. Applied to the five toenail models from earlier, QIC ranks independence as most adequate, closely followed by Toeplitz, AR(1), and exchangeable, with unstructured penalized for its larger number of nuisance parameters.

| Working Correlation | QIC               |
|---------------------|-------------------|
| Independence        | 1833.34 (lowest)  |
| Toeplitz            | 1833.97           |
| AR(1)               | 1834.18           |
| Exchangeable        | 1834.38           |
| Unstructured        | 1836.26 (highest) |

## CHOOSING AND CHECKING A MODEL, 3 OF 3

### An Honest Tension, Not a Clean Answer

The chapter is candid about something genuinely intriguing in its own results: QIC points to independence as the most adequate working correlation for the toenail data, despite the concerns raised earlier about how inefficient independence can be for testing the treatment effect.

Meanwhile, unstructured — the very structure QIC penalizes most, for its large number of nuisance parameters — is the one that produces the smallest, most clearly significant robust p-value (0.0285) for the treatment effect. A single number cannot settle which working correlation to trust; QIC, the robust-versus-naïve comparison, and substantive judgment about the data all belong in the decision together.

## UNBALANCED DESIGNS, 1 OF 3

# When There Is No Natural Order At All

Observational data are often unbalanced in a different way than the toenail trial's dropout pattern: sometimes there is no natural ordering among the repeated measures to begin with, or the ordering is not aligned across subjects. A National Toxicology Program developmental toxicity study, reported by Molenberghs and Verbeke (2005), illustrates this well.

Di(2-ethylhexyl)phthalate (DEHP) is used in vacuum pumps and as a plasticizer; small amounts can leak from PVC plastic containers in the presence of food, milk, blood, or solvents, raising a genuine human health question. Tyl et al. (1988) exposed pregnant mice to daily DEHP doses of 0, 44, 91, 191, or 292 mg/kg/day, sacrificed the dams before delivery, and examined the contents of the uterus — a total of 1,082 live fetuses.

Here, fetuses are clustered within dams (litters), with no meaningful order among littermates — a genuinely different clustering structure from the toenail trial's ordered visits over time.

## UNBALANCED DESIGNS, 2 OF 3

# Modeling the Chance of Malformation

The outcome of interest is the “collapsed” malformation indicator: whether a fetus had at least one of three malformation types — external, visceral, or skeletal. A logistic regression models the probability of malformation as a function of dose, rescaled from 0 (placebo) to 1 (the highest dose, 292 mg/kg/day).

Because littermates cannot be meaningfully ordered, only independence or exchangeable working correlations make sense here — AR(1), Toeplitz, and unstructured all require an ordering that simply does not exist among fetuses in the same litter.

Starting from independence, GEE1, Prentice’s method, and the linearization-based method all agree closely: an intercept of about -3.98 (SE 0.30) and a dose effect of about 5.56 (SE 0.61), using empirically corrected standard errors.

UNBALANCED DESIGNS, 3 OF 3

Five Methods, One Consistent Conclusion

Under an exchangeable working correlation, all five GEE-family methods agree: dose has a significant effect on malformation risk, with closely similar point and precision estimates.

| Method        | Intercept    | Dose Effect | Association             |
|---------------|--------------|-------------|-------------------------|
| Standard GEE1 | -4.05 (0.31) | 5.84 (0.61) | correlation 0.11        |
| Prentice      | -4.06 (0.31) | 5.89 (0.61) | correlation 0.15 (0.05) |
| Linearization | -4.04 (0.31) | 5.82 (0.61) | correlation 0.11        |
| GEE2          | -5.23 (0.40) | 5.35 (0.60) | correlation 0.09 (0.01) |
| ALR           | -4.04 (0.31) | 5.93 (0.63) | odds ratio 3.22 (0.93)  |

The exchangeable association is modest but significant throughout — and, because GEE2 uses the information more efficiently than Prentice's method, its correlation estimate carries a noticeably smaller standard error (0.01 versus 0.05). ALR's odds ratio and the other methods' correlations sit on different scales, so they cannot be compared number-for-number, but all point the same way: a real, if modest, degree of association among littermates.

## CONCLUDING REMARKS, 1 OF 4

# A Practical Answer for a Common Problem

GEE is a widely used semi-parametric approach for modeling longitudinal and hierarchical data when responses are not necessarily Gaussian, and scientific interest lies in the marginal, population-averaged mean. Random-effects models are usually less suitable for this purpose, since their fixed-effects parameters typically lack a direct marginal interpretation — and fully specified marginal models quickly become unwieldy as the number of repeated measurements grows.

This deck has walked through Liang and Zeger's original 1986 proposal and its major alternatives — Prentice's method, GEE2, alternating logistic regression, and a linearization-based approach — illustrated on a two-armed longitudinal trial with a binary outcome, and extended to an unbalanced, unordered clustering problem in a developmental toxicity study.

GEE is not restricted to binary or count outcomes: it can also be used with Gaussian data when robust marginal inference — rather than a subject-specific interpretation — is the analytic goal.

## CONCLUDING REMARKS, 2 OF 4

### A Major Limitation: Missing Data

A drawback of GEE is its lack of a likelihood or Bayesian foundation — a gap that matters most when data are incomplete, as they were for roughly a quarter of the toenail trial’s patients. It is one of several limitations the chapter discusses, alongside efficiency loss, model assessment, and the added computational burden of some extensions.

Likelihood and Bayesian methods stay valid under the comparatively mild “missing at random” (MAR) assumption: whether data are missing can depend on what was already observed, as long as it does not further depend on the unobserved values themselves. Frequentist methods like GEE need something stronger.

## CONCLUDING REMARKS, 3 OF 4

# Why This Matters, and How It Is Being Fixed

Plain GEE needs the much stronger “missing completely at random” (MCAR) assumption: missingness must be independent of both observed and unobserved responses, depending only on covariates. In practice this is a real limitation — a patient dropping out of a trial because of how their treatment is going is exactly the kind of missingness MCAR rules out.

Two lines of research have relaxed this requirement: inverse probability weighting, leading to **weighted GEE** (WGEE), and **multiple imputation combined with GEE** (MI-GEE). Both let GEE-based analyses tolerate the weaker, more realistic MAR assumption instead.

## Right About What Matters, Even When Wrong About the Rest

For standard GEE1, a correctly specified mean model protects the marginal regression estimates even when the working correlation is wrong — the reassurance this whole chapter is built around. Extensions that model the association itself, such as GEE2, trade some of that protection away: they additionally require the association model to be correctly specified.